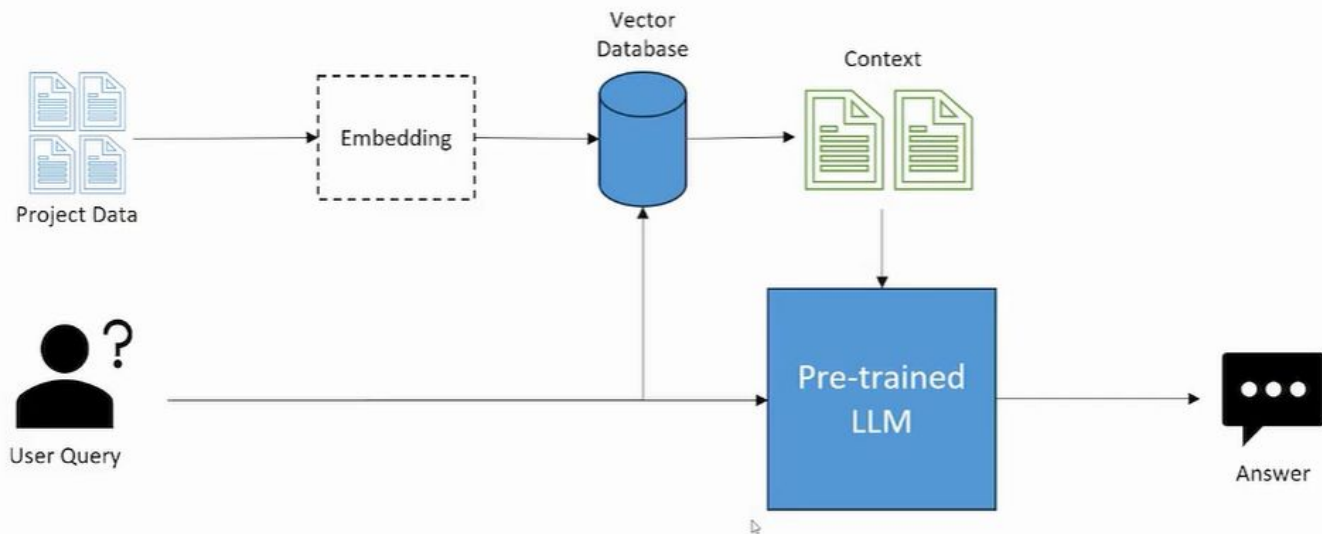




IN-CONTEXT LEARNING WITH PERSONAL DATA

Using Llama2 and FAISS



LARGE LANGUAGE MODEL



VECTOR DATABASES



LARGE LANGUAGE MODELS

- What is an LLM?
- How LLMs work?
- What are their applications?
- Importance of LLMs in the project.

VECTOR DATABASES

- What is a vector database?
- Their role in organizing and retrieving vectorized data
- Benefits of using a vector database with LLMs.



- Neural networks - Brains behind the AI
- Tokenization - Chopping up language
- Encoding - Turning words into numbers

LARGE LANGUAGE
MODELS
IN DEPTH

THE MAGIC OF THE TRANSFORMER

- Self-attention - The AI's focus lens
- Why it's awesome - Speed, smarts, and understanding
- Real-world magic - Translating, summarizing, chatting away

TRAINING AND FINE-TUNING LLMS

- Building a brainy base with pre-training
- Fine-tuning - Making it extra sharp for special tasks
- The need for powerhouse computers
- The never-ending quest for smarter AI



OVERVIEW OF LLAMA 2

- 2 trillion "tokens" from diverse sources
- Bytepair encoding (BPE) for tokenization
- transformer architecture with advanced features

PRE-TRAINING AND FINE- TUNING

- Vast initial training on 2 trillion tokens
- Supervised Fine-Tuning (SFT) for model refinement
- Reinforcement Learning with Human Feedback (RLHF)
- Customization for specific needs (e.g., brand voice)



REINFORCEMENT LEARNING WITH HUMAN FEEDBACK (RLHF)

- Detailed annotator guidance in response evaluation.
- Employment of Proximal Policy Optimization (PPO).
- Use of Rejection Sampling Fine-Tuning for model improvement.
- Development of safety and helpfulness reward models



CUSTOMIZATION AND OPEN INNOVATION

- Llama 2 supports extensive customization options
- Open-source model with detailed documentation
- Compatibility with cloud platforms for scalability
- Restrictions for large tech companies ensure wide accessibility

VECTOR DATABASES IN DEPTH

- Store data as vectors in high-dimensional space
- Key for efficient similarity searches in machine learning applications
- Distinction from traditional relational databases
- Applications - Recommendation engines, image retrieval, natural language processing

VECTORS IN AI

- Vectors as numerical representations of complex data
- Embedding processes - Converting text, images, and more into vectors
- Dimensionality and its impact on data representation
- Real-world examples - Facial recognition, language translation

VECTOR DATABASES IN SEARCH

- Accelerating similarity searches in large datasets
- Principle of similarity search: Nearest neighbor algorithms
- Applications in personalization and content discovery
- Enhancing user experiences through accurate, fast retrieval



FAISS

- A library for efficient similarity search and clustering of dense vectors
- Developed by Facebook AI Research for large-scale vector search
- Features - High performance, scalability, and support for various indexing strategies
- Components - Index objects, quantization, and GPU acceleration

HOW FAISS WORKS

- Indexing vectors for quick retrieval
- Trade-offs between search speed and accuracy
- Quantization - Reducing vector size for storage efficiency
- Integration capabilities with Python and popular machine learning frameworks



ADVANCED FEATURES OF FAISS

- Support for different types of indexes for specific use cases
- Customizable parameters for fine-tuning search performance
- Clustering capabilities for organizing large datasets
- Extensive documentation and community support